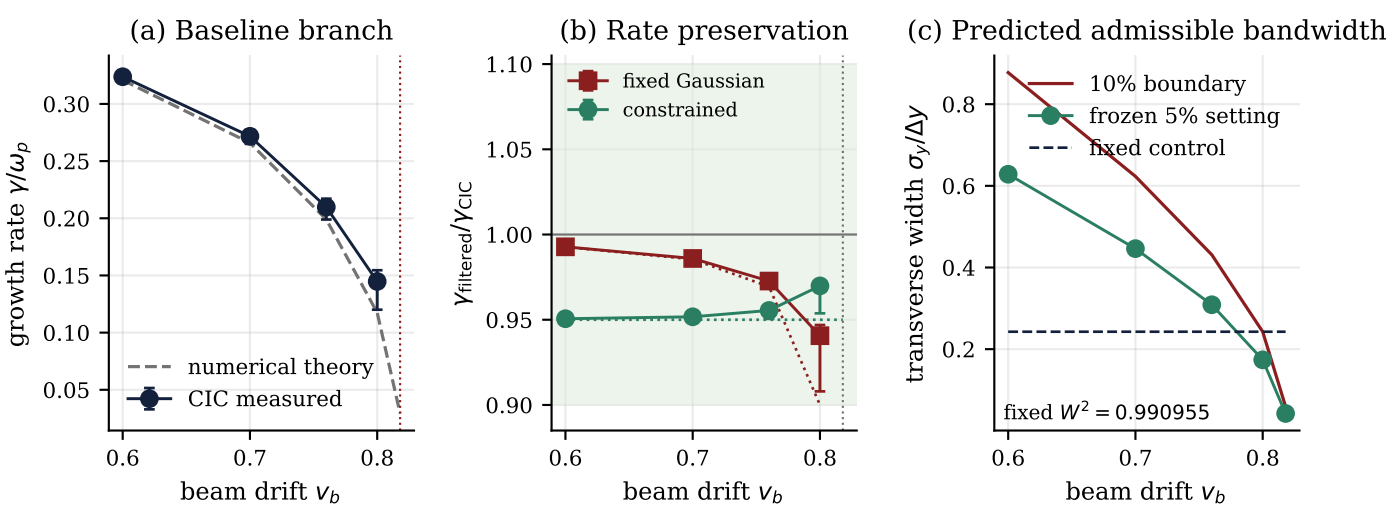
(a) Baseline branch
growth rate γ/ω_p
0.30
0.25
0.20
0.15
0.10
0.05
numerical theory
CIC measured
0.6
0.7
0.8
beam drift v_b
(b) Rate preservation
$\gamma_{\mathrm{filtered}}/\gamma_{\mathrm{CIC}}$
1.10
1.05
1.00
0.95
0.90
fixed Gaussian
constrained
0.6
0.7
0.8
beam drift v_b
(c) Predicted admissible bandwidth
transverse width $\sigma_y/\Delta y$
0.8
0.6
0.4
0.2
10% boundary
frozen 5% setting
fixed control
fixed $W^2 = 0.990955$
0.6
0.7
0.8
beam drift v_b